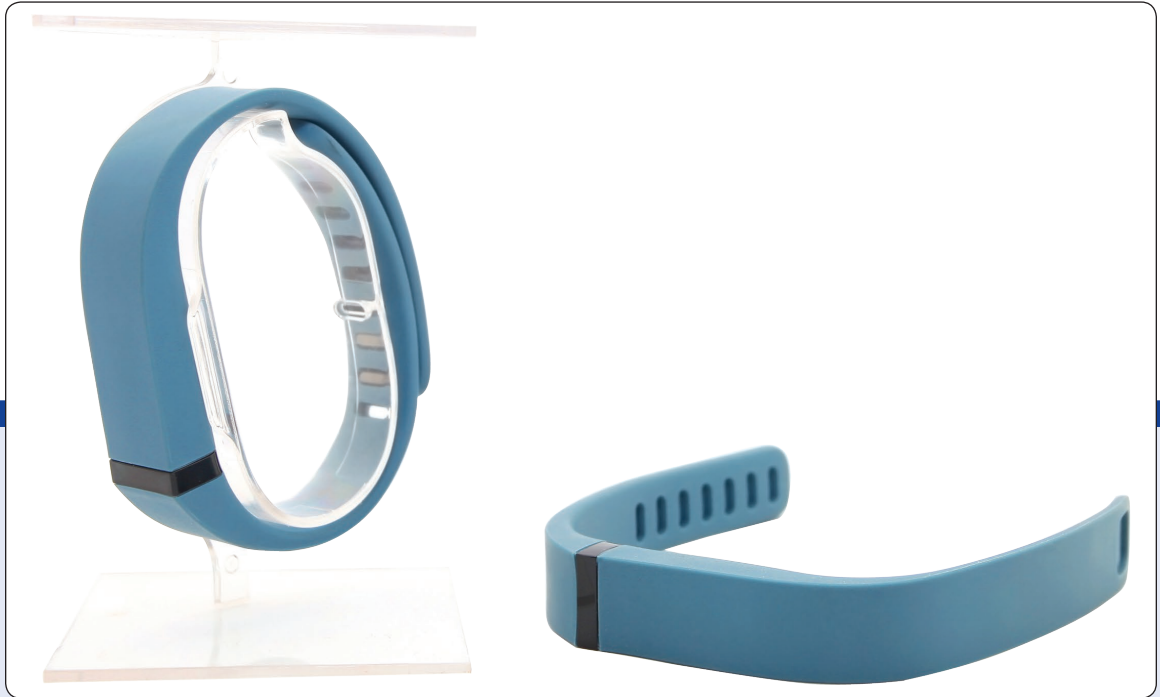


What is Statistics?



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- ▲ **BEST BUY** sells Fitbit wearable technology products that track a person's physical activity and sleep quality. The Fitbit technology collects daily information on a person's number of steps so that a person can track calories consumed. The information can be synced with a cell phone and displayed with a Fitbit app. Assume you know the daily number of Fitbit Flex 2 units sold last month at the Best Buy store in Collegeville, Pennsylvania. Describe a situation where the number of units sold is considered a sample. Illustrate a second situation where the number of units sold is considered a population. (See Exercise 11 and **LO1-3**.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

- LO1-1** Explain why knowledge of statistics is important.
- LO1-2** Define statistics and provide an example of how statistics is applied.
- LO1-3** Differentiate between descriptive and inferential statistics.
- LO1-4** Classify variables as qualitative or quantitative, and discrete or continuous.
- LO1-5** Distinguish between nominal, ordinal, interval, and ratio levels of measurement.
- LO1-6** List the values associated with the practice of statistics.



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INTRODUCTION

Suppose you work for a large company and your supervisor asks you to decide if a new version of a smartphone should be produced and sold. You start by thinking about the product's innovations and new features. Then, you stop and realize the consequences of the decision. The product will need to make a profit so the pricing and the costs of production and distribution are all very important. The decision to introduce the product is based on many alternatives. So how will you know? Where do you start?

Without a long experience in the industry, beginning to develop an intelligence that will make you an expert is essential. You select three other people to work with and meet with them. The conversation focuses on what you need to know and what information and data you need. In your meeting, many questions are asked. How many competitors are already in the market? How are smartphones priced? What design features do competitors' products have? What features does the market require? What do customers want in a smartphone? What do customers like about the existing products? The answers will be based on business intelligence consisting of data and information collected through customer surveys, engineering analysis, and market research. In the end, your presentation to support your decision regarding the introduction of a new smartphone is based on the statistics that you use to summarize and organize your data, the statistics that you use to compare the new product to existing products, and the statistics to estimate future sales, costs, and revenues. The statistics will be the focus of the conversation that you will have with your supervisor about this very important decision.

As a decision maker, you will need to acquire and analyze data to support your decisions. The purpose of this text is to develop your knowledge of basic statistical techniques and methods and how to apply them to develop the business and personal intelligence that will help you make decisions.

LO1-1

Explain why knowledge of statistics is important.

WHY STUDY STATISTICS?

If you look through your university catalogue, you will find that statistics is required for many college programs. As you investigate a future career in accounting, economics, human resources, finance, business analytics, or other business area, you also will discover that statistics is required as part of these college programs. So why is statistics a requirement in so many disciplines?

A major driver of the requirement for statistics knowledge is the technologies available for capturing data. Examples include the technology that Google uses to track how Internet users access websites. As people use Google to search the Internet, Google records every search and then uses these data to sort and prioritize the results for future Internet searches. One recent estimate indicates that Google processes 20,000 terabytes of information per day. Big-box retailers like Target, Walmart, Kroger, and others scan every purchase and use the data to manage the distribution of products, to make decisions about marketing and sales, and to track daily and even hourly sales. Police departments collect and use data to provide city residents with maps that communicate information about crimes committed and their location. Every organization is collecting and using data to develop knowledge and intelligence that will help people make informed decisions, and to track the implementation of their decisions. The graphic to the left shows the amount of data generated every minute (www.domo.com). A good working knowledge of statistics is useful for summarizing and organizing data to provide information that is useful and supportive of decision making. Statistics is used to make valid comparisons and to predict the outcomes of decisions.

In summary, there are at least three reasons for studying statistics: (1) data are collected everywhere and require statistical knowledge to



make the information useful, (2) statistical techniques are used to make professional and personal decisions, and (3) no matter what your career, you will need a knowledge of statistics to understand the world and to be conversant in your career. An understanding of statistics and statistical method will help you make more effective personal and professional decisions.

LO1-2

Define statistics and provide an example of how statistics is applied.

STATISTICS IN ACTION

A feature of our textbook is called *Statistics in Action*. Read each one carefully to get an appreciation of the wide application of statistics in management, economics, nursing, law enforcement, sports, and other disciplines.

- In 2015, *Forbes* published a list of the richest Americans. William Gates, founder of Microsoft Corporation, is the richest. His net worth is estimated at \$76.0 billion. (www.forbes.com)
- In 2015, the four largest privately owned American companies, ranked by revenue, were Cargill, Koch Industries, Dell, and Albertsons. (www.forbes.com)
- In the United States, a typical high school graduate earns \$668 per week, a typical college graduate with a bachelor's degree earns \$1,101 per week, and a typical college graduate with a master's degree earns \$1,326 per week. (www.bls.gov/emp/ep_chart_001.htm)

WHAT IS MEANT BY STATISTICS?

This question can be rephrased in two, subtly different ways: what are statistics and what is statistics? To answer the first question, a statistic is a number used to communicate a piece of information. Examples of **statistics** are:

- The inflation rate is 2%.
- Your grade point average is 3.5.
- The price of a new Tesla Model S sedan is \$79,570.

Each of these statistics is a numerical fact and communicates a very limited piece of information that is not very useful by itself. However, if we recognize that each of these statistics is part of a larger discussion, then the question “what **is** statistics” is applicable. Statistics is the set of knowledge and skills used to organize, summarize, and analyze data. The results of statistical analysis will start interesting conversations in the search for knowledge and intelligence that will help us make decisions. For example:

- The inflation rate for the calendar year was 0.7%. By applying statistics we could compare this year's inflation rate to the past observations of inflation. Is it higher, lower, or about the same? Is there a trend of increasing or decreasing inflation? Is there a relationship between interest rates and government bonds?
- Your grade point average (GPA) is 3.5. By collecting data and applying statistics, you can determine the required GPA to be admitted to the Master of Business Administration program at the University of Chicago, Harvard, or the University of Michigan. You can determine the likelihood that you would be admitted to a particular program. You may be interested in interviewing for a management position with Procter & Gamble. What GPA does Procter & Gamble require for college graduates with a bachelor's degree? Is there a range of acceptable GPAs?
- You are budgeting for a new car. You would like to own an electric car with a small carbon footprint. The price for the Tesla Model S Sedan is \$79,570. By collecting additional data and applying statistics, you can analyze the alternatives. For example, another choice is a hybrid car that runs on both gas and electricity such as a 2015 Toyota Prius. It can be purchased for about \$28,659. Another hybrid, the Chevrolet Volt, costs \$33,995. What are the differences in the cars' specifications? What additional information can be collected and summarized so that you can make a good purchase decision?

Another example of using statistics to provide information to evaluate decisions is the distribution and market share of Frito-Lay products. Data are collected on each of the Frito-Lay product lines. These data include the market share and the pounds of product sold. Statistics is used to present this information in a bar chart in Chart 1–1. It clearly shows Frito-Lay's dominance in the potato, corn, and tortilla chip markets. It also shows the absolute measure of pounds of each product line consumed in the United States.

These examples show that statistics is more than the presentation of numerical information. Statistics is about collecting and processing information to create a conversation, to stimulate additional questions, and to provide a basis for making decisions. Specifically, we define **statistics** as:

STATISTICS The science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making more effective decisions.

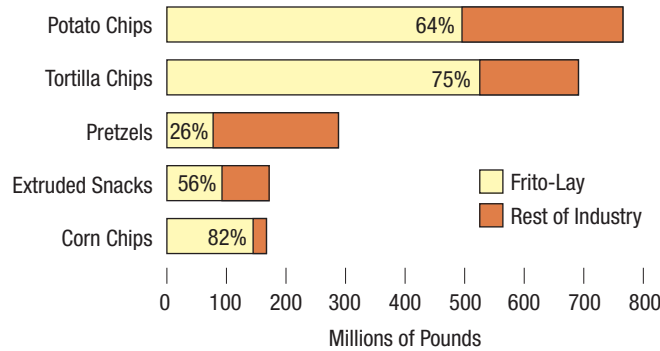


CHART 1–1 Frito-Lay Volume and Share of Major Snack Chip Categories in U.S. Supermarkets

In this book, you will learn the basic techniques and applications of statistics that you can use to support your decisions, both personal and professional. To start, we will differentiate between descriptive and inferential statistics.

LO1-3

Differentiate between descriptive and inferential statistics.

TYPES OF STATISTICS

When we use statistics to generate information for decision making from data, we use either descriptive statistics or inferential statistics. Their application depends on the questions asked and the type of data available.

Descriptive Statistics

Masses of unorganized data—such as the census of population, the weekly earnings of thousands of computer programmers, and the individual responses of 2,000 registered voters regarding their choice for president of the United States—are of little value as is. However, descriptive statistics can be used to organize data into a meaningful form. We define **descriptive statistics** as:

DESCRIPTIVE STATISTICS Methods of organizing, summarizing, and presenting data in an informative way.

The following are examples that apply descriptive statistics to summarize a large amount of data and provide information that is easy to understand.

- There are a total of 46,837 miles of interstate highways in the United States. The interstate system represents only 1% of the nation's total roads but carries more than 20% of the traffic. The longest is I-90, which stretches from Boston to Seattle, a distance of 3,099 miles. The shortest is I-878 in New York City, which is 0.70 mile in length. Alaska does not have any interstate highways, Texas has the most interstate miles at 3,232, and New York has the most interstate routes with 28.
- The average person spent \$133.91 on traditional Valentine's Day merchandise in 2014. This is an increase of \$2.94 from 2013. As in previous years, men spent more than twice the amount women spent on the holiday. The average man spent \$108.38 to impress the people in his life while women only spent \$48.41.

Statistical methods and techniques to generate descriptive statistics are presented in Chapters 2 and 4. These include organizing and summarizing data with frequency distributions and presenting frequency distributions with charts and graphs. In addition, statistical measures to summarize the characteristics of a distribution are discussed in Chapter 3.

Inferential Statistics

Sometimes we must make decisions based on a limited set of data. For example, we would like to know the operating characteristics, such as fuel efficiency measured by miles per gallon, of sport utility vehicles (SUVs) currently in use. If we spent a lot of time, money, and effort, all the owners of SUVs could be surveyed. In this case, our goal would be to survey the **population** of SUV owners.

POPULATION The entire set of individuals or objects of interest or the measurements obtained from all individuals or objects of interest.

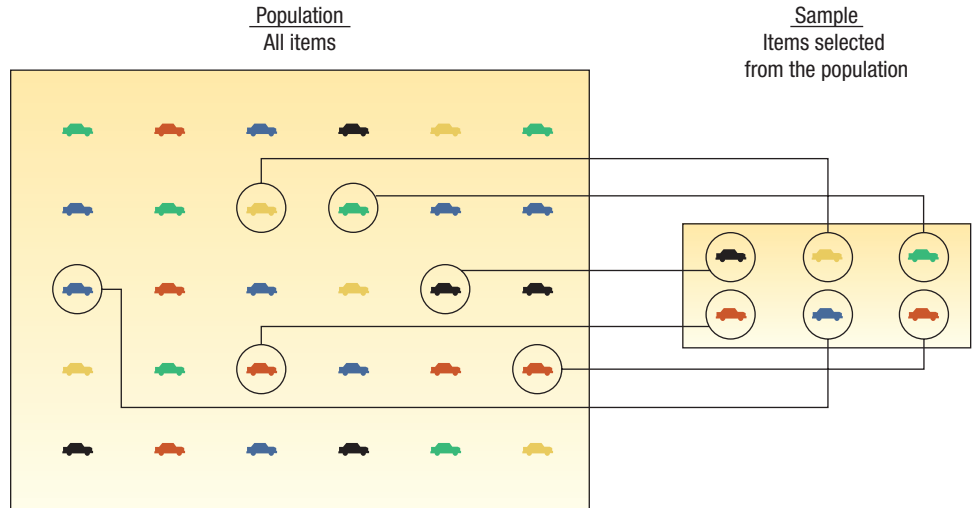
However, based on inferential statistics, we can survey a limited number of SUV owners and collect a **sample** from the population.

SAMPLE A portion, or part, of the population of interest.

Samples often are used to obtain reliable estimates of population parameters. (Sampling is discussed in Chapter 8.) In the process, we make trade-offs between the time, money, and effort to collect the data and the error of estimating a population parameter. The process of sampling SUVs is illustrated in the following graphic. In this example, we would like to know the mean or average SUV fuel efficiency. To estimate the mean of the population, six SUVs are sampled and the mean of their MPG is calculated.

STATISTICS IN ACTION

Where did statistics get its start? In 1662 John Graunt published an article called “Natural and Political Observations Made upon Bills of Mortality.” The author’s “observations” were the result of a study and analysis of a weekly church publication called “Bill of Mortality,” which listed births, christenings, and deaths and their causes. Graunt realized that the Bills of Mortality represented only a fraction of all births and deaths in London. However, he used the data to reach broad conclusions or inferences about the impact of disease, such as the plague, on the general population. His logic is an example of statistical inference. His analysis and interpretation of the data are thought to mark the start of statistics.



So, the sample of six SUVs represents evidence from the population that we use to reach an inference or conclusion about the average MPG for all SUVs. The process of sampling from a population with the objective of estimating properties of a population is called **inferential statistics**.

INFERENTIAL STATISTICS The methods used to estimate a property of a population on the basis of a sample.

Inferential statistics is widely applied to learn something about a population in business, agriculture, politics, and government, as shown in the following examples:

- Television networks constantly monitor the popularity of their programs by hiring Nielsen and other organizations to sample the preferences of TV viewers. For example, 9.0% of a sample of households with TVs watched The Big Bang Theory during the week of November 2, 2015 (www.nielsen.com). These program ratings are used to make decisions about advertising rates and whether to continue or cancel a program.
- In 2015, a sample of U.S. Internal Revenue Service tax preparation volunteers were tested with three standard tax returns. The sample indicated that tax returns were completed with a 49% accuracy rate. In other words there were errors on about half of the returns. In this example, the statistics are used to make decisions about how to improve the accuracy rate by correcting the most common errors and improving the training of volunteers.

A feature of our text is self-review problems. There are a number of them interspersed throughout each chapter. The first self-review follows. Each self-review tests your comprehension of preceding material. The answer and method of solution are given in Appendix E. You can find the answer to the following self-review in 1–1 in Appendix E. We recommend that you solve each one and then check your answer.

SELF-REVIEW 1–1



The answers are in Appendix E.

The Atlanta-based advertising firm Brandon and Associates asked a sample of 1,960 consumers to try a newly developed chicken dinner by Boston Market. Of the 1,960 sampled, 1,176 said they would purchase the dinner if it is marketed.

- Is this an example of descriptive statistics or inferential statistics? Explain.
- What could Brandon and Associates report to Boston Market regarding acceptance of the chicken dinner in the population?

LO1-4

Classify variables as qualitative or quantitative, and discrete or continuous.

TYPES OF VARIABLES

There are two basic types of variables: (1) qualitative and (2) quantitative (see Chart 1–2). When an object or individual is observed and recorded as a nonnumeric characteristic, it is a qualitative variable or an attribute. Examples of qualitative variables are gender, beverage preference, type of vehicle owned, state of birth, and eye color. When a variable is quantitative, we usually count the number of observations for each category and determine

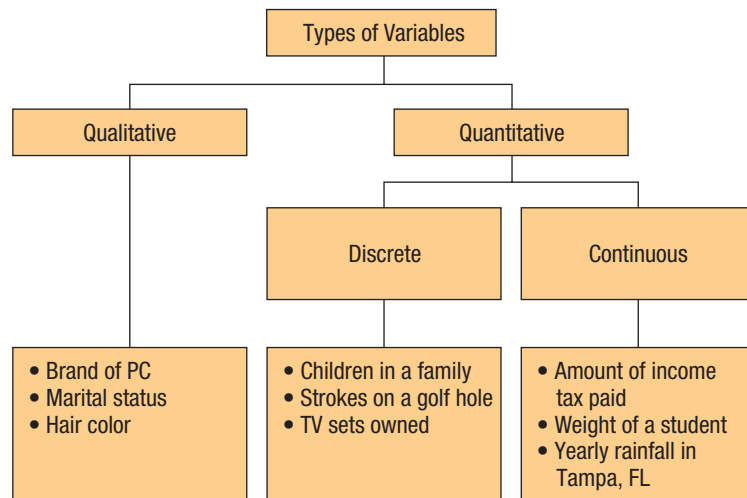


CHART 1–2 Summary of the Types of Variables

what percent are in each category. For example, if we observe the variable eye color, what percent of the population has blue eyes and what percent has brown eyes? If the variable is type of vehicle, what percent of the total number of cars sold last month were SUVs? Qualitative variables are often summarized in charts and bar graphs (Chapter 2).

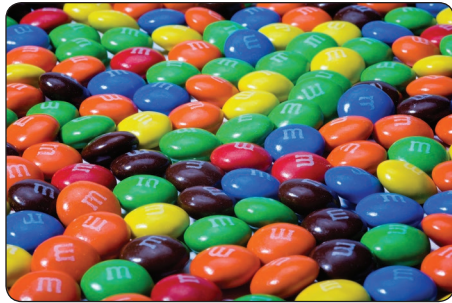
When a variable can be reported numerically, it is called a quantitative variable. Examples of quantitative variables are the balance in your checking account, the number of gigabytes of data used on your cell phone plan last month, the life of a car battery (such as 42 months), and the number of people employed by a company.

Quantitative variables are either discrete or continuous. Discrete variables can assume only certain values, and there are “gaps” between the values. Examples of discrete variables are the number of bedrooms in a house (1, 2, 3, 4, etc.), the number of cars arriving at Exit 25 on I-4 in Florida near Walt Disney World in an hour (326, 421, etc.), and the number of students in each section of a statistics course (25 in section A, 42 in section B, and 18 in section C). We count, for example, the number of cars arriving at Exit 25 on I-4, and we count the number of statistics students in each section. Notice that a home can have 3 or 4 bedrooms, but it cannot have 3.56 bedrooms. Thus, there is a “gap” between possible values. Typically, discrete variables are counted.

Observations of a continuous variable can assume any value within a specific range. Examples of continuous variables are the air pressure in a tire and the weight of a shipment of tomatoes. Other examples are the ounces of raisins in a box of raisin bran cereal and the duration of flights from Orlando to San Diego. Grade point average (GPA) is a continuous variable. We could report the GPA of a particular student as 3.2576952. The usual practice is to round to 3 places—3.258. Typically, continuous variables result from measuring.

LO1-5

Distinguish between nominal, ordinal, interval, and ratio levels of measurement.



© Ron Buskirk/Alamy Stock Photo

LEVELS OF MEASUREMENT

Data can be classified according to levels of measurement. The level of measurement determines how data should be summarized and presented. It also will indicate the type of statistical analysis that can be performed. Here are two examples of the relationship between measurement and how we apply statistics. There are six colors of candies in a bag of M&Ms. Suppose we assign brown a value of 1, yellow 2, blue 3, orange 4, green 5, and red 6. What kind of variable is the color of an M&M? It is a qualitative variable. Suppose someone summarizes M&M color by adding the assigned color values, divides the sum by the number of M&Ms, and reports that the mean color is 3.56. How do we interpret this statistic? You are correct in concluding that it has no meaning as a measure of M&M color. As a qualitative variable, we can only report the count and percentage of each color in a bag of M&Ms. As a second example, in a high school track meet there are eight competitors in the 400-meter run. We report the order of finish and that the mean finish is 4.5. What does the mean finish tell us? Nothing! In both of these instances, we have not used the appropriate statistics for the level of measurement.

There are four levels of measurement: nominal, ordinal, interval, and ratio. The lowest, or the most primitive, measurement is the nominal level. The highest is the ratio level of measurement.

Nominal-Level Data

For the **nominal level of measurement**, observations of a qualitative variable are measured and recorded as labels or names. The labels or names can only be classified and counted. There is no particular order to the labels.

NOMINAL LEVEL OF MEASUREMENT Data recorded at the nominal level of measurement is represented as labels or names. They have no order. They can only be classified and counted.

The classification of the six colors of M&M milk chocolate candies is an example of the nominal level of measurement. We simply classify the candies by color. There is no natural order. That is, we could report the brown candies first, the orange first, or any of the other colors first. Recording the variable gender is another example of the nominal level of measurement. Suppose we count the number of students entering a football game with a student ID and report how many are men and how many are women. We could report either the men or the women first. For the data measured at the nominal level, we are limited to counting the number in each category of the variable. Often, we convert these counts to percentages. For example, a random sample of M&M candies reports the following percentages for each color:

Color	Percent in a bag
Blue	24%
Green	20%
Orange	16%
Yellow	14%
Red	13%
Brown	13%

To process the data for a variable measured at the nominal level, we often numerically code the labels or names. For example, if we are interested in measuring the home state for students at East Carolina University, we would assign a student's home state of Alabama a code of 1, Alaska a code of 2, Arizona a 3, and so on. Using this procedure with an alphabetical listing of states, Wisconsin is coded 49 and Wyoming 50. Realize that the number assigned to each state is still a label or name. The reason we assign numerical codes is to facilitate counting the number of students from each state with statistical software. Note that assigning numbers to the states does not give us license to manipulate the codes as numerical information. Specifically, in this example, $1 + 2 = 3$ corresponds to Alabama + Alaska = Arizona. Clearly, the nominal level of measurement does not permit any mathematical operation that has any valid interpretation.

Ordinal-Level Data

The next higher level of measurement is the **ordinal level**. For this level of measurement a qualitative variable or attribute is either ranked or rated on a relative scale.

ORDINAL LEVEL OF MEASUREMENT Data recorded at the ordinal level of measurement is based on a relative ranking or rating of items based on a defined attribute or qualitative variable. Variables based on this level of measurement are only ranked or counted.

Best Business Climate
1. Florida
2. Utah
3. Texas
4. Georgia
5. Indiana
6. Tennessee
7. Nebraska
8. North Carolina
9. Virginia
10. Washington

For example, many businesses make decisions about where to locate their facilities; in other words, where is the best place for their business? Business Facilities (www.businessfacilities.com) publishes a list of the top 10 states for the “best business climate.” The 2016 rankings are shown to the left. They are based on the evaluation of many different factors, including the cost of labor, business tax climate, quality of life, transportation infrastructure, educated workforce, and economic growth potential.

This is an example of an ordinal scale because the states are ranked in order of best to worst business climate. That is, we know the relative order of the states based

on the attribute. For example, in 2016 Florida had the best business climate and Utah was second. Indiana was fifth, and that was better than Tennessee but not as good as Georgia. Notice we cannot say that Florida's business climate is five times better than Indiana's business climate because the magnitude of the differences between the states is not known. To put it another way, we do not know if the magnitude of the difference between Louisiana and Utah is the same as between Texas and Georgia.

Another example of the ordinal level measure is based on a scale that measures an attribute. This type of scale is used when students rate instructors on a variety of attributes. One attribute may be: "Overall, how do you rate the quality of instruction in this class?" A student's response is recorded on a relative scale of inferior, poor, good, excellent, and superior. An important characteristic of using a relative measurement scale is that we cannot distinguish the magnitude of the differences between groups. We do not know if the difference between "Superior" and "Good" is the same as the difference between "Poor" and "Inferior."

Table 1–1 lists the frequencies of 60 student ratings of instructional quality for Professor James Brunner in an Introduction to Finance course. The data are summarized based on the order of the scale used to rate the instructor. That is, they are summarized by the number of students who indicated a rating of superior (6), good (26), and so on. We also can convert the frequencies to percentages. About 43.3% (26/60) of the students rated the instructor as good.

TABLE 1–1 Rating of a Finance Professor

Rating	Frequency	Percentage
Superior	6	10.0%
Good	26	43.3%
Average	16	26.7%
Poor	9	15.0%
Inferior	3	5.0%

Interval-Level Data

The **interval level of measurement** is the next highest level. It includes all the characteristics of the ordinal level, but, in addition, the difference or interval between values is meaningful.

INTERVAL LEVEL OF MEASUREMENT For data recorded at the interval level of measurement, the interval or the distance between values is meaningful. The interval level of measurement is based on a scale with a known unit of measurement.

The Fahrenheit temperature scale is an example of the interval level of measurement. Suppose the high temperatures on three consecutive winter days in Boston are 28, 31, and 20 degrees Fahrenheit. These temperatures can be easily ranked, but we can also determine the interval or distance between temperatures. This is possible because 1 degree Fahrenheit represents a constant unit of measurement. That is, the distance between 10 and 15 degrees Fahrenheit is 5 degrees, and is the same as the 5-degree distance between 50 and 55 degrees Fahrenheit. It is also important to note that 0 is just a point on the scale. It does not represent the absence of the condition. The measurement of zero degrees Fahrenheit does not represent the absence of heat or cold. But by our own measurement scale, it is cold! A major limitation of a variable measured at the interval level is that we cannot make statements similar to 20 degrees Fahrenheit is twice as warm as 10 degrees Fahrenheit.

Another example of the interval scale of measurement is women’s dress sizes. Listed below is information on several dimensions of a standard U.S. woman’s dress.

Size	Bust (in)	Waist (in)	Hips (in)
8	32	24	35
10	34	26	37
12	36	28	39
14	38	30	41
16	40	32	43
18	42	34	45
20	44	36	47
22	46	38	49
24	48	40	51
26	50	42	53
28	52	44	55

Why is the “size” scale an interval measurement? Observe that as the size changes by two units (say from size 10 to size 12 or from size 24 to size 26), each of the measurements increases by 2 inches. To put it another way, the intervals are the same. There is no natural zero point for dress size. A “size 0” dress does not have “zero” material. Instead, it would have a 24-inch bust, 16-inch waist, and 27-inch hips. Moreover, the ratios are not reasonable. If you divide a size 28 by a size 14, you do not get the same answer as dividing a size 20 by a size 10. Neither ratio is equal to two, as the “size” number would suggest. In short, if the distances between the numbers make sense, but the ratios do not, then you have an interval scale of measurement.

Ratio-Level Data

Almost all quantitative variables are recorded on the **ratio level of measurement**. The ratio level is the “highest” level of measurement. It has all the characteristics of the interval level, but, in addition, the 0 point and the ratio between two numbers are both meaningful.

RATIO LEVEL OF MEASUREMENT Data recorded at the ratio level of measurement are based on a scale with a known unit of measurement and a meaningful interpretation of zero on the scale.

Examples of the ratio scale of measurement include wages, units of production, weight, changes in stock prices, distance between branch offices, and height. Money is also a good illustration. If you have zero dollars, then you have no money, and a wage of \$50 per hour is two times the wage of \$25 per hour. Weight also is measured at the ratio level of measurement. If a scale is correctly calibrated, then it will read 0 when nothing is on the scale. Further, something that weighs 1 pound is half as heavy as something that weighs 2 pounds.

Table 1–2 illustrates the ratio scale of measurement for the variable, annual income for four father-and-son combinations. Observe that the senior Lahey earns twice as much as his son. In the Rho family, the son makes twice as much as the father.

TABLE 1–2 Father–Son Income Combinations

Name	Father	Son
Lahey	\$80,000	\$ 40,000
Nale	90,000	30,000
Rho	60,000	120,000
Steele	75,000	130,000

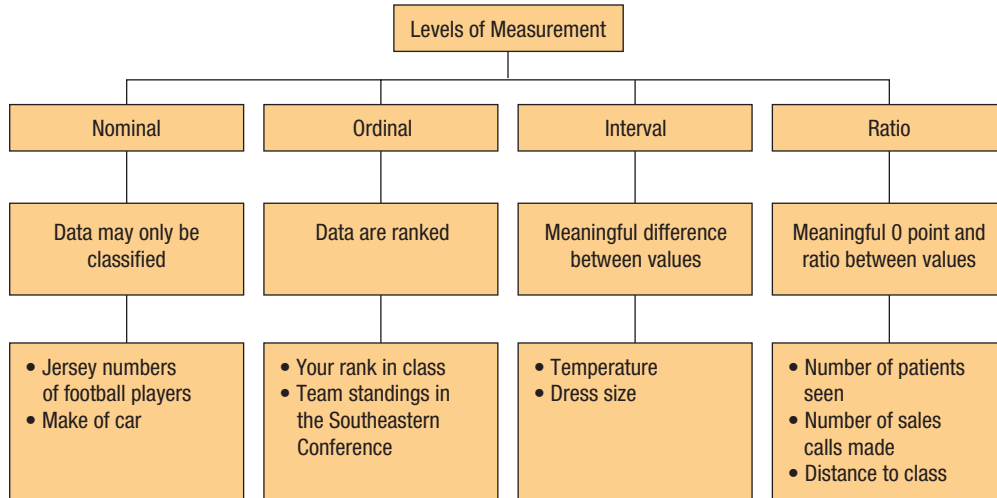


CHART 1–3 Summary and Examples of the Characteristics for Levels of Measurement

Chart 1–3 summarizes the major characteristics of the various levels of measurement. The level of measurement will determine the type of statistical methods that can be used to analyze a variable. Statistical methods to analyze variables measured on a nominal level are discussed in Chapter 15; methods for ordinal-level variables are discussed in Chapter 16. Statistical methods to analyze variables measured on an interval or ratio level are presented in Chapters 9 through 14.

SELF-REVIEW 1–2



- The mean age of people who listen to talk radio is 42.1 years. What level of measurement is used to assess the variable age?
- In a survey of luxury-car owners, 8% of the U.S. population owned luxury cars. In California and Georgia, 14% of people owned luxury cars. Two variables are included in this information. What are they and how are they measured?

EXERCISES

The answers to the odd-numbered exercises are in Appendix D.

- What is the level of measurement for each of the following variables?
 - Student IQ ratings.
 - Distance students travel to class.
 - The jersey numbers of a sorority soccer team.
 - A student's state of birth.
 - A student's academic class—that is, freshman, sophomore, junior, or senior.
 - Number of hours students study per week.
- Slate* is a daily magazine on the Web. Its business activities can be described by a number of variables. What is the level of measurement for each of the following variables?
 - The number of hits on their website on Saturday between 8:00 am and 9:00 am.
 - The departments, such as food and drink, politics, foreign policy, sports, etc.
 - The number of weekly hits on the Sam's Club ad.
 - The number of years each employee has been employed with *Slate*.
- On the Web, go to your favorite news source and find examples of each type of variable. Write a brief memo that lists the variables and describes them in terms of qualitative or quantitative, discrete or continuous, and the measurement level.

4. For each of the following, determine whether the group is a sample or a population.
 - a. The participants in a study of a new cholesterol drug.
 - b. The drivers who received a speeding ticket in Kansas City last month.
 - c. People on welfare in Cook County (Chicago), Illinois.
 - d. The 30 stocks that make up the Dow Jones Industrial Average.

LO1-6

List the values associated with the practice of statistics.

ETHICS AND STATISTICS

Following events such as Wall Street money manager Bernie Madoff's Ponzi scheme, which swindled billions from investors, and financial misrepresentations by Enron and Tyco, business students need to understand that these events were based on the misrepresentation of business and financial information. In each case, people within each organization reported financial information to investors that indicated the companies were performing much better than the actual situation. When the true financial information was reported, the companies were worth much less than advertised. The result was many investors lost all or nearly all of the money they had invested.

The article "Statistics and Ethics: Some Advice for Young Statisticians," in *The American Statistician* 57, no. 1 (2003), offers guidance. The authors advise us to practice statistics with integrity and honesty, and urge us to "do the right thing" when collecting, organizing, summarizing, analyzing, and interpreting numerical information. The real contribution of statistics to society is a moral one. Financial analysts need to provide information that truly reflects a company's performance so as not to mislead individual investors. Information regarding product defects that may be harmful to people must be analyzed and reported with integrity and honesty. The authors of *The American Statistician* article further indicate that when we practice statistics, we need to maintain "an independent and principled point-of-view" when analyzing and reporting findings and results.

As you progress through this text, we will highlight ethical issues in the collection, analysis, presentation, and interpretation of statistical information. We also hope that, as you learn about using statistics, you will become a more informed consumer of information. For example, you will question a report based on data that do not fairly represent the population, a report that does not include all relevant statistics, one that includes an incorrect choice of statistical measures, or a presentation that introduces bias in a deliberate attempt to mislead or misrepresent.

BASIC BUSINESS ANALYTICS

A knowledge of statistics is necessary to support the increasing need for companies and organizations to apply business analytics. Business analytics is used to process and analyze data and information to support a story or narrative of a company's business, such as "what makes us profitable," "how will our customers respond to a change in marketing"? In addition to statistics, an ability to use computer software to summarize, organize, analyze, and present the findings of statistical analysis is essential. In this text, we will be using very elementary applications of business analytics using common and available computer software. Throughout our text, we will use Microsoft Excel and, occasionally, Minitab. Universities and colleges usually offer access to Microsoft Excel. Your computer already may be packaged with Microsoft Excel. If not, the Microsoft Office package with Excel often is sold at a reduced academic price through your university or college. In this text, we use Excel for the majority of the applications. We also use an Excel "Add-in" called MegaStat. If your instructor requires this package, it is available at www.mhhe.com/megastat. This add-in gives Excel the capability to produce additional statistical reports. Occasionally, we use Minitab to illustrate an application. See www.minitab.com for further information. Minitab also offers discounted academic pricing. The 2016 version of Microsoft Excel supports the analyses in our text. However,

earlier versions of Excel for Apple Mac computers do not have the necessary add-in. If you do not have Excel 2016 and are using an Apple Mac computer with Excel, you can download the free, trial version of Stat Plus at www.analystsoft.com. It is a statistical software package that will integrate with Excel for Mac computers.

The following example shows the application of Excel to perform a statistical summary. It refers to sales information from the Applewood Auto Group, a multi-location car sales and service company. The Applewood information has sales information for 180 vehicle sales. Each sale is described by several variables: the age of the buyer, whether the buyer is a repeat customer, the location of the dealership for the sale, the type of vehicle sold, and the profit for the sale. The following shows Excel's summary of statistics for the variable profit. The summary of profit shows the mean profit per vehicle was \$1,843.17, the median profit was slightly more at \$1,882.50, and profit ranged from \$294 to \$3,292.

APPLEWOOD AUTO GROUP.xlsx							
	A	B	C	D	E	F	G
1	Age	Profit	Location	Vehicle-Type	Previous		Profit
2	33	\$1,889	Olean	SUV	1		
3	47	\$1,461	Kane	Sedan	0	Mean	1843.17
4	44	\$1,532	Tionesta	SUV	3	Standard Error	47.97
5	53	\$1,220	Olean	Sedan	0	Median	1882.50
6	51	\$1,674	Sheffield	Sedan	1	Mode	1915.00
7	41	\$2,389	Kane	Truck	1	Standard Deviation	643.63
8	58	\$2,058	Kane	SUV	1	Sample Variance	414256.61
9	35	\$1,919	Tionesta	SUV	1	Kurtosis	-0.22
10	45	\$1,266	Olean	Sedan	0	Skewness	-0.24
11	54	\$2,991	Tionesta	Sedan	0	Range	2998
12	56	\$2,695	Kane	Sedan	2	Minimum	294
13	41	\$2,165	Tionesta	SUV	0	Maximum	3292
14	38	\$1,766	Sheffield	SUV	0	Sum	331770
15	48	\$1,952	Tionesta	Compact	1	Count	180

Throughout the text, we will motivate the use of computer software to summarize, describe, and present information and data. The applications of Excel are supported by instructions so that you can learn how to apply Excel to do statistical analysis. The instructions are presented in Appendix C of this text. These data and other data sets and files are available on the text's student website, www.mhhe.com/lind17e.

CHAPTER SUMMARY

- I. Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making more effective decisions.
- II. There are two types of statistics.
 - A. Descriptive statistics are procedures used to organize and summarize data.
 - B. Inferential statistics involve taking a sample from a population and making estimates about a population based on the sample results.
 1. A population is an entire set of individuals or objects of interest or the measurements obtained from all individuals or objects of interest.
 2. A sample is a part of the population.
- III. There are two types of variables.
 - A. A qualitative variable is nonnumeric.
 1. Usually we are interested in the number or percent of the observations in each category.
 2. Qualitative data are usually summarized in graphs and bar charts.

- B. There are two types of quantitative variables and they are usually reported numerically.
 - 1. Discrete variables can assume only certain values, and there are usually gaps between values.
 - 2. A continuous variable can assume any value within a specified range.
- IV. There are four levels of measurement.
 - A. With the nominal level, the data are sorted into categories with no particular order to the categories.
 - B. The ordinal level of measurement presumes that one classification is ranked higher than another.
 - C. The interval level of measurement has the ranking characteristic of the ordinal level of measurement plus the characteristic that the distance between values is a constant size.
 - D. The ratio level of measurement has all the characteristics of the interval level, plus there is a 0 point and the ratio of two values is meaningful.

CHAPTER EXERCISES

- 5. Explain the difference between qualitative and quantitative variables. Give an example of qualitative and quantitative variables.
- 6. Explain the difference between a sample and a population.
- 7. Explain the difference between a discrete and a continuous variable. Give an example of each not included in the text.
- 8. For the following situations, would you collect information using a sample or a population? Why?
 - a. Statistics 201 is a course taught at a university. Professor Rauch has taught nearly 1,500 students in the course over the past 5 years. You would like to know the average grade for the course.
 - b. As part of a research project, you need to report the average profit as a percentage of revenue for the #1-ranked corporation in the Fortune 500 for each of the last 10 years.
 - c. You are looking forward to graduation and your first job as a salesperson for one of five large pharmaceutical corporations. Planning for your interviews, you will need to know about each company's mission, profitability, products, and markets.
 - d. You are shopping for a new MP3 music player such as the Apple iPod. The manufacturers advertise the number of music tracks that can be stored in the memory. Usually, the advertisers assume relatively short, popular songs to estimate the number of tracks that can be stored. You, however, like Broadway musical tunes and they are much longer. You would like to estimate how many Broadway tunes will fit on your MP3 player.
- 9. Exits along interstate highways were formerly numbered successively from the western or southern border of a state. However, the Department of Transportation has recently changed most of them to agree with the numbers on the mile markers along the highway.
 - a. What level of measurement were data on the consecutive exit numbers?
 - b. What level of measurement are data on the milepost numbers?
 - c. Discuss the advantages of the newer system.
- 10. A poll solicits a large number of college undergraduates for information on the following variables: the name of their cell phone provider (AT&T, Verizon, and so on), the numbers of minutes used last month (200, 400, for example), and their satisfaction with the service (Terrible, Adequate, Excellent, and so forth). What is the level of measurement for each of these three variables?
- 11. Best Buy sells Fitbit wearable technology products that track a person's activity. For example, the Fitbit technology collects daily information on a person's number of steps so that a person can track calories consumed. The information can be synced with a cell phone and displayed with a Fitbit app. Assume you know the daily number of Fitbit Flex

2 units sold last month at the Best Buy store in Collegeville, Pennsylvania. Describe a situation where the number of units sold is considered a sample. Illustrate a second situation where the number of units sold is considered a population.

12. Using the concepts of sample and population, describe how a presidential election is unlike an “exit” poll of the electorate.
13. Place these variables in the following classification tables. For each table, summarize your observations and evaluate if the results are generally true. For example, salary is reported as a continuous quantitative variable. It is also a continuous ratio-scaled variable.
 - a. Salary
 - b. Gender
 - c. Sales volume of MP3 players
 - d. Soft drink preference
 - e. Temperature
 - f. SAT scores
 - g. Student rank in class
 - h. Rating of a finance professor
 - i. Number of home video screens

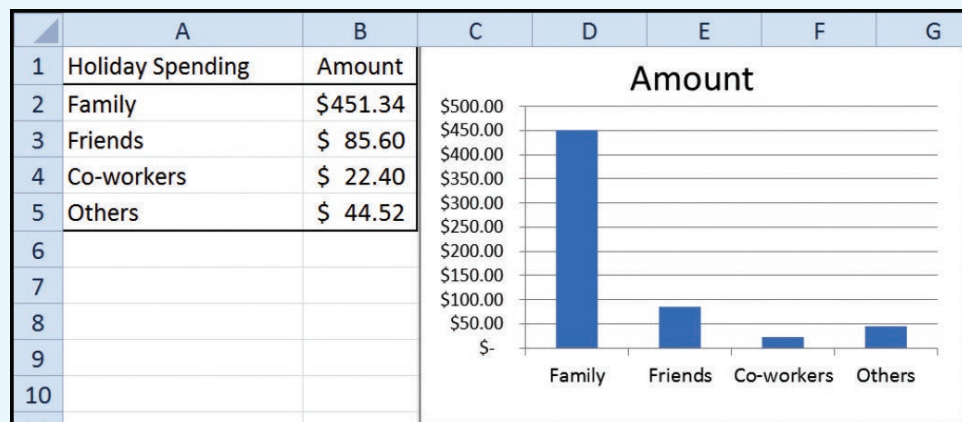
	Discrete Variable	Continuous Variable
Qualitative		
Quantitative		a. Salary

	Discrete	Continuous
Nominal		
Ordinal		
Interval		
Ratio		a. Salary

14. Using data from such publications as the *Statistical Abstract of the United States*, *Forbes*, or any news source, give examples of variables measured with nominal, ordinal, interval, and ratio scales.
15. The Struthers Wells Corporation employs more than 10,000 white-collar workers in its sales offices and manufacturing facilities in the United States, Europe, and Asia. A sample of 300 U.S. workers revealed 120 would accept a transfer to a location outside the United States. On the basis of these findings, write a brief memo to Ms. Wanda Carter, Vice President of Human Services, regarding all white-collar workers in the firm and their willingness to relocate.
16. AVX Home Entertainment, Inc., recently began a “no-hassles” return policy. A sample of 500 customers who recently returned items showed 400 thought the policy was fair, 32 thought it took too long to complete the transaction, and the rest had no opinion. On the basis of this information, make an inference about customer reaction to the new policy.
17. **FILE** The *Wall Street Journal*’s website, www.wsj.com, reported the number of cars and light-duty trucks sold through October of 2014 and October of 2015. The top sixteen manufacturers are listed here. Sales data often is reported in this way to compare current sales to last year’s sales.

Manufacturer	Year-to-Date Sales	
	Through October 2015	Through October 2014
General Motors Corp.	2,562,840	2,434,707
Ford Motor Company	2,178,587	2,065,612
Toyota Motor Sales USA Inc.	2,071,446	1,975,368
Chrysler	1,814,268	1,687,313
American Honda Motor Co Inc.	1,320,217	1,281,777
Nissan North America Inc.	1,238,535	1,166,389
Hyundai Motor America	638,195	607,539
Kia Motors America Inc.	526,024	489,711
Subaru of America Inc.	480,331	418,497
Volkswagen of America Inc.	294,602	301,187
Mercedes-Benz	301,915	281,728
BMW of North America Inc.	279,395	267,193
Mazda Motor of America Inc.	267,158	259,751
Audi of America Inc.	165,103	146,133
Mitsubishi Motors N A, Inc.	80,683	64,564
Volvo	53,803	47,823

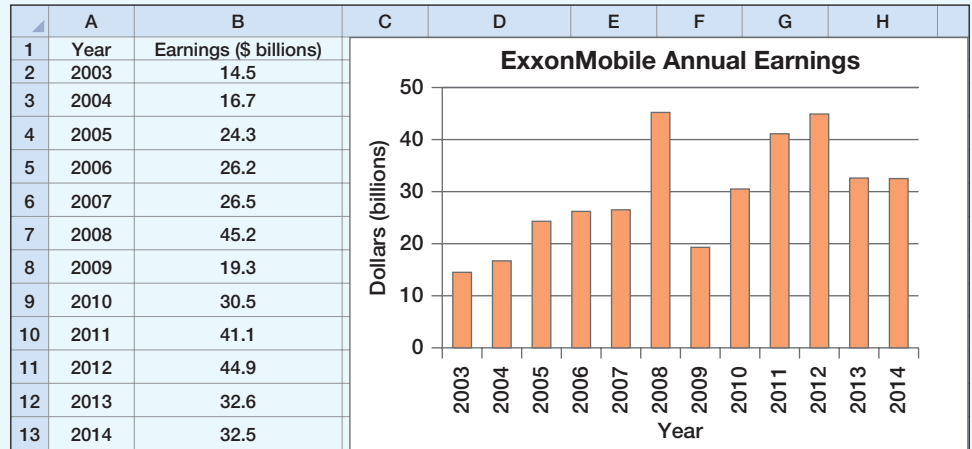
- Using computer software, compare the October 2015 sales to the October 2014 sales for each manufacturer by computing the difference. Make a list of the manufacturers that increased sales compared to 2014; make a list of manufacturers that decreased sales.
 - Using computer software, compare 2014 sales to 2015 sales for each manufacturer by computing the percentage change in sales. Make a list of the manufacturers in order of increasing percentage changes. Which manufacturers are in the top five in percentage change? Which manufacturers are in the bottom five in percentage change?
 - Using computer software, first sort the data using the 2015 year-to-date sales. Then, design a bar graph to illustrate the 2014 and 2015 year-to-date sales for the top 12 manufacturers. Also, design a bar graph to illustrate the percentage change for the top 12 manufacturers. Compare these two graphs and prepare brief written comments.
18. The following chart depicts the average amounts spent by consumers on holiday gifts.



Write a brief report summarizing the amounts spent during the holidays. Be sure to include the total amount spent and the percent spent by each group.

19. The following chart depicts the earnings in billions of dollars for ExxonMobil for the period 2003 until 2014. Write a brief report discussing the earnings at ExxonMobil during

the period. Was one year higher than the others? Did the earnings increase, decrease, or stay the same over the period?



DATA ANALYTICS

20. **FILE** Refer to the North Valley Real Estate data, which report information on homes sold in the area last year. Consider the following variables: selling price, number of bedrooms, township, and mortgage type.
 - a. Which of the variables are qualitative and which are quantitative?
 - b. How is each variable measured? Determine the level of measurement for each of the variables.
21. **FILE** Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season. Consider the following variables: number of wins, payroll, season attendance, whether the team is in the American or National League, and the number of home runs hit.
 - a. Which of these variables are quantitative and which are qualitative?
 - b. Determine the level of measurement for each of the variables.
22. **FILE** Refer to the Lincolnville School District bus data, which report information on the school district's bus fleet.
 - a. Which of the variables are qualitative and which are quantitative?
 - b. Determine the level of measurement for each variable.